Independent Operations, Dependent Histories

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Abstract

A finite network can admit every combination of local operations and still admit only one history for each combination. The quantifiers are different. One ranges over a product of operation types. The other ranges over assignments of event values. An invented three-workshop case makes the split concrete, and an ordinary serial circuit shows that constrained event values are not, by themselves, a failure of metaphysical recombination. A reversible extension of the same finite rules does not convert operation-substitution into free recombination of concrete objects, and it does not establish a physical implementation. The positive proposal is modest: an implementation claim must say which components and operation types vary, which combinations of substitutions are admitted, and which physical responses are supposed to match the formal histories.

1 Two freedoms

The following network is invented. It is a mathematical description, not a machine anyone is asked to build.

Three workshops each receive one bit and emit one bit. At each workshop the allowed operations are all four functions from a bit to a bit. Nothing in the model forbids any of those functions, and nothing forbids any triple of them. There are sixty-four such triples. A global rule then sends the triple of emitted bits to the triple of received bits. For each choice of local functions, the equations of the network have exactly one solution. Call a solution a history: a triple of emitted bits that, sent through the rule and then through the chosen functions, returns itself.

Two modal remarks are then easy to run together. The first says that the local operations are independent, because every combination of them is admitted. The second says that the histories are independent, because the bits emitted at the three workshops can be recombined freely. The second remark is false of this network. For a fixed triple of functions, most assignments of

emitted bits are not histories. The first remark can still be true. The operations that may be installed are not the event values that may jointly occur once the operations are installed.

The two inferences to refuse can be written before any theorem is used. First: every combination of local operations is admitted; therefore every combination of emitted bits is a possible history. Second: not every combination of emitted bits is a history; therefore the local operations are not independent, or the workshops are not modally free concrete things. The premises of the first can be true while the conclusion is false, because the admitted items are functions and the history is a value-tuple. The premise of the second can be true while both conclusions are false, because a constraint on value-tuples is not a deletion from the menu of functions and is not a fact about concrete natures. The rest of the paper is the demonstration that the three-workshop equations realize the first pattern, that a serial circuit realizes it too, and that neither pattern is Schaffer's recombination or Chalmers's implementation.

That is the thesis. Independence of permitted local operations is not independence of the events that make up a computational history. A model can include the full product of allowed operation types and, for each factor in that product, only one globally constrained history. Any modal claim about the model has to say which partition and which intervention class it is using. Neither the success of operation substitution nor the failure of arbitrary event recombination, without further premises, settles ontological independence or physical implementation.

The positive proposal is a constraint on how implementation claims are assessed, not a definition of computation. Call it a typed substitution principle. An implementation claim must identify which components and which operation types are held fixed or varied, which combinations of those substitutions are admitted, and what correspondence between formal histories and physical responses is being asserted. The principle is compatible with a causal or counterfactual account of implementation. It does not replace one.

An ordinary serial circuit will serve as a control, introduced in the next section. It also forbids many assignments of wire values. Constrained histories are therefore not a mark of metaphysical exoticism. What is special about the three workshops is not that their event values are constrained. It is that the constraint coexists with a full product of local operations and with no fixed order among the three slots.

The mathematical claims below are reported in an unpublished technical note (Arrowood, unpublished). Related drafts of that note are already public, so this manuscript does not claim that anonymity is secure. No address is given. A neutral statement of assistance appears in Section 8. Placement of that statement relative to anonymous review is unresolved.

2 Operations and histories

Fix the types before drawing morals. There are three parties, $k \in \{1, 2, 3\}$, each with input alphabet $I_k = \{0, 1\}$ and output alphabet $O_k = \{0, 1\}$. Let F_k be the set of all functions from I_k to O_k. Each F_k has four members. A substitution is an element of the product

$$F = F_1 \times F_2 \times F_3,$$

which has sixty-four elements. Write $O = O_1 \times O_2 \times O_3$, the set of eight possible output triples. These eight triples are the candidate event values. They are not the substitutions.

A rule is a function $w : O \to I$, with $I = I_1 \times I_2 \times I_3$. Given a substitution $f = (f_1, f_2, f_3)$ and a rule w, a history for that pair is an output triple o such that

$$o = f(w(o)),$$

where f acts componentwise. Let $H(w, f)$ be the set of such histories. The rule w is a process function when

$$(\forall f \in F)\, |H(w, f)| = 1$$

(Baumeler and Wolf, 2016). The universal quantifier ranges over substitutions, not over output triples. For each substitution the history set is required to be a singleton. It is not required to be all of O.

The running example is the three-bit rule

$$w(a, b, c) = (\neg b \wedge c,\ \neg c \wedge a,\ \neg a \wedge b)$$

(Baumeler and Wolf, 2016; Baumeler et al., 2019). Each input depends on the other two outputs. There is no ordering of the three slots in which each depends only on earlier slots. One substitution can be checked by hand. Let each f_k be the identity. Then a history must satisfy $a = \neg b \wedge c$, $b = \neg c \wedge a$, and $c = \neg a \wedge b$. If $a = 1$, then $b = 0$ and $c = 1$, and the third equation says $c = 0$. If $a = 0$, then $b = 0$ and $c = 0$, and $(0, 0, 0)$ works. Exactly one history. A second substitution shows that the unique history is not a label pasted onto every choice. Let each f_k ignore its input and write 1. The only output triple those functions can emit is $(1, 1, 1)$. The rule sends $(1, 1, 1)$ to $(0, 0, 0)$, and the constant functions send $(0, 0, 0)$ back to $(1, 1, 1)$. So $(1, 1, 1)$ is a history, and nothing else can be, because nothing else is in the image of f. The identity substitution and the constant substitution are both in F, and they have different histories. Operation independence is the joint admission of the two substitutions. History dependence is the fact that each of them solves for one triple and not for the other seven. This

case does not prove the universal claim. It shows the shape of $H(w, f)$. The claim that every one of the sixty-four substitutions has exactly one history is part of the standing example in the unpublished note. It is not re-enumerated here.

Two propositions are compatible because their variables differ.

> **Operation independence.** $(\forall f \in F)$ the substitution f is included in the domain of the theory. No factor of the product is struck out.
>
> **History dependence.** $(\forall f \in F)(\exists! o \in O)\, o \in H(w, f)$. For each admitted substitution, exactly one output triple is a history.

The cardinalities make the split hard to miss, on the assumption, taken from the standing example and not re-proved here, that every substitution has exactly one history. There are 64 substitutions and 8 output triples, hence 512 pairs (f, o). History dependence says that exactly 64 of those pairs are solutions, one per substitution. Operation independence says that all 64 substitutions occur as first components of some pair in the domain. Event recombination would say something much stronger: for each f, all 8 values of o are solutions, hence all 512 pairs are solutions. The network, if the standing example is right, hits the first two counts and misses the third. A reader who hears "64" and "free combination" has heard the count of substitutions. A reader who hears "only one triple works" has heard the count of histories. They are not hearing the same number, and they are not hearing a count of metaphysically possible worlds. Worlds are not pairs (f, o).

Operation independence does not say that every $o \in O$ is a history for a given f. History dependence does not say that some f is excluded. A third proposition, which the example rejects, would be event recombination: for a fixed f, every output triple is a history, or any way of setting the three emitted bits can be combined with any other. That proposition quantifies over O, or over ways the event values can be, inside one substitution. The process-function clause quantifies over F and then constrains $H(w, f)$.

Zero histories and several histories are different failures, and neither is the rejection of a substitution from F. If $H(w, f)$ is empty, the chosen functions and the rule cannot be satisfied together. If $H(w, f)$ has two or more members, the chosen functions have not determined which history occurs. Multiple solutions are underdetermination, not a contradiction and not a metaphysical impossibility. The process-function policy excludes both failures by requiring a singleton. It is a determination policy for histories, given that every substitution remains available.

The control case is also invented, and it is ordinary. Consider two gates in series. The first copies its input or flips it; both settings are allowed. The second does the same. Every pair of

gate settings is admitted. Not every assignment of values to the two wires satisfies the gates. If both gates are set to copy, the two wires must carry the same bit. The forbidden assignments are not evidence that the gates cannot be set independently, and they are not evidence that the wires are a single metaphysical substance. They are evidence that, once the settings are fixed, the event values have to solve the circuit equations. An acyclic computer already separates operation settings from wire values. The three-workshop network uses the same separation. Its additional feature is the absence of a fixed order among the slots, not the bare fact that some value-assignments fail.

A serial feedback loop should not be recruited as the same phenomenon. A thermostat's diagram may draw a circular arrow. Unrolled in time, the history is a sequence of states of the same devices: cool room, furnace on; warm room, furnace off. The later state is not an input to the earlier state inside one simultaneous triple of slots. The three workshops are not three times. They are three operation slots whose inputs may depend on one another's outputs with no slot earliest in the dependence. Calling both arrangements "loops" does not identify their quantifiers.

3 A reversible test

The same split survives a change of intervention class, and the change has to be named. Deterministic functions in F are one class. Local unitaries on finite ancillas are another. They are not a notation for each other.

For any rule w, process function or not, there is a permutation of a finite basis that uses a source system and a sink system,

$$U|o\rangle|e\rangle = |w(o) + e\rangle|o\rangle,$$

with addition in each input alphabet, once that alphabet is read as a cyclic group of the same order. A permutation of a finite basis is unitary. Bare unitarity of U therefore does not distinguish process functions, and it is not a physical realization. The ledger exists for rules with no history and for rules with many.

The nontrivial test inserts local unitaries. For each party, let V_k be a unitary from $I_k \otimes X_k$ to $O_k \otimes Y_k$, with finite ancillas satisfying $|I_k|\,|X_k| = |O_k|\,|Y_k|$. Wiring those unitaries into U defines a closed loop M. The unpublished note shows that M is unitary for every such choice if and only if w is a process function. The forward direction uses exclusivity. If w is a process function and two output strings differ, some party has a different output symbol and the same input symbol on those strings. Otherwise one substitution would have both strings as histories. The

Gram matrix of M factorizes over parties, and that party's factor vanishes, so distinct rows are orthogonal. Matched dimensions make M square, hence unitary. The converse uses a different gadget. If some substitution has $N \neq 1$ histories, those functions can be copied into unitaries on a larger space, and a column of M then has squared norm N.

The equivalence is for this canonical extension and this family of tests. It is not the claim that an arbitrary dilation is unitary exactly when w is a process function, and it is not the claim that purification in any wider sense coincides with the fixed-point clause. A local unitary that is not diagonal in the computational basis mixes basis states the classical table treated as distinct. Survival of unitarity under all such mixings is a result about process functions, not a rewriting of the definition. The ancillas are part of the test. The incoming state may still entangle them. Entanglement is a feature of the state, not an extra hypothesis on w. Instruments that are not unitary are a further class again. A local completely positive map on a finite system can be dilated to a unitary on a larger finite space, after which the environment is discarded. That dilation is how one passes from the unitary tests to a larger family of operations. It is not an extra identification of those operations with metaphysical ways a thing can be. An assessor who treats "we varied the local operation" as one gesture, equally good for a bit-function, a unitary, an instrument, and a change in a concrete object's nature, has used one English phrase for four quantifiers. The reversible test speaks only to the unitary quantifier, and only for the canonical U.

What the equivalence does, for the modal point, is limited and real. Moving from bit-functions to all matched local unitaries does not, in this finite construction, add a further rejection of process functions. The operation-substitution condition and the unitary-robustness condition draw the same line. That is a robustness result inside one formalism. It does not say that varying a formal function is varying a concrete object's intrinsic nature, location, or existence. It does not supply a Hamiltonian, a region of spacetime, or a protocol for setting a physical part. Vilasini and Renner (2024) give no-go results about embedding cyclic information-theoretic structures in acyclic spacetimes. The equivalence is not an exception to those results, because it is not an embedding.

One boundary is enough to mark the finite hypothesis.[1] A known-board aside belongs here and nowhere else. In the computational basis, with a stated source, one pure process induces a valid classical distribution on three binary parties that lies outside the convex hull of the process functions. The separating value reported in the unpublished note is $\frac{1}{4} - \frac{3}{16}\sqrt{2} < 0$. The board is not presented as new, and the separation is not claimed for every basis. It blocks one

[1]With infinitely many parties, pairwise exclusivity still yields at most one fixed point, while existence can fail. Whether every exclusive map of Baire class one has a fixed point is open. The arguments of this paper use finite parties and finite alphabets only.

reading of one record: that this distribution is ignorance over process functions. It is not a theory of recombination.

4 Recombination

Schaffer (2010) reconstructs an argument from internal relatedness to priority monism. The reconstruction is his, and the historical authors he discusses are not re-read here. What is used is his own apparatus, because it is a careful account of what metaphysical independence would have to be, and because a cyclic diagram is easy to mistake for an instance of that apparatus.

Two preliminaries keep the comparison from being a slogan. First, the monism at issue is not the view that only the world exists. It is priority monism: a claim about the dependence order among concrete objects, on which the whole is prior and its proper parts are posterior. Schaffer is explicit that the priority in question is not the causal notion on which an effect depends on a cause, and not the conceptual notion on which an analysandum depends on an analysans. It is metaphysical dependence for nature and existence. A functional equation among bit values is none of those three relations until a bridge premise says that it is. Second, he treats free recombination as the modal signature of a disconnected plurality, and failure of free recombination as the modal signature of an integrated whole. In §4 he leaves open at least two pluralist replies, one of which defends free recombination and one of which posits brute modality. He does not claim to have closed those replies. Nothing below is offered as a counterexample to his proofs. The workshops do not instantiate his assumptions.

His working picture of modal freedom is the stereo. There are no necessary connections between the settings of two knobs. Any way one knob can be set, and any way the other can be set, is a way both can be set. All combinations are possible. In §1.4 he sharpens a necessary condition. If two entities are modally free, then for any way the first can be and any way the second can be, there is a metaphysically possible world in which each is that way, barring co-location and leaving the rest of the world as it is. The ways, in his unpacking, are intrinsic natures, spatiotemporal locations, and the one way of failing to exist. The world that realizes a combination is supposed to leave the rest of the world as it is, barring co-location. That last clause matters. A constraint that merely says which bit-triples solve an equation, while remaining silent about every other object, is not yet a constraint on how the rest of a world may be. Schaffer also notes that he is stating a necessary condition clear enough for later use, not a finished analysis of free recombination, and that Lewis's coexistence gloss is a different notion, one that does not settle whether the one thing must be altered to coexist with the other. The comparison below uses only the necessary condition he states. It does not borrow a quotation from the historical authors he cites, and it does not treat his reading of them as a reading of

their books. An internal relation, in the constraining sense he isolates, is one that precludes this freedom for its relata.

Apply that template to the workshops and watch the quantifiers miss.

The product F does look like a stereo. Any function in F_1, any function in F_2, and any function in F_3 may be chosen together. If "way a workshop can be" meant "which operation type is installed," the model would be exhibiting Schaffer's knob combination for operation types. The model does not say that. The operation types are formal objects in a product of function sets. They are not intrinsic natures of concrete workshops, not locations, and not existence or nonexistence. Calling the product modal freedom in Schaffer's sense adds a premise the equations do not contain: that each formal function is a way a concrete object can be, in the how-where-whether list.

The history constraint looks, just as misleadingly, like a failure of recombination. For the identity substitution, $(1, 1, 1)$ is not a history. Not every combination of emitted bits occurs. If "way a workshop can be" meant "which bit it emits in this substitution," the model would be constraining combinations of event values. That constraint is real, and it is the same kind of constraint the serial circuit imposes on its wires. Schaffer's failure of free recombination is a constraint on natures, locations, and existence across metaphysically possible worlds. A fixed-point equation is a constraint on which value-tuples solve a function for a fixed substitution. The serial control shows that the second constraint is commonplace. It does not, by itself, mean that the three workshops fail to be modally free concrete objects, because the workshops have not been given natures, locations, or a world-relative existence.

The bridge premises that would be required, and that are not supplied, are these. First, each workshop is a concrete object with an intrinsic nature, a location, and a possible absence, and the relevant ways it can be are those. Second, installing a local function, or emitting a bit, is identical with or grounds one of those ways. Third, the fixed-point restriction holds with metaphysical necessity for those objects, not merely as a clause in a model. Fourth, the dependence recorded by w is Schaffer's irreflexive, transitive, well-founded ontological dependence, rather than causal dependence or a bare functional equation. Any one of these can be denied while the process function stands. Denying them is not a criticism of Schaffer. It is a refusal to pretend that his theorem has been fed.

Even a successful bridge would not be a refutation of pluralism. Schaffer's §4 leaves the pluralist two open replies: defend free recombination, and so reject the assumption that the constraining relations hold universally; or posit brute modality, and so reject the assumptions that tie the modal constraints to a ground in the categorical. He says he has no knock-down argument against either reply. A Boolean network does not supply one. If the bridge premises were granted, the network would be one more alleged example of modal constraint, to be

weighed with the others he discusses. If they are not granted, the network is not such an example at all. Either way, priority monism is not a corollary of $|H(w, f)| = 1$, and the denial of priority monism is not a corollary either. The paper's comparison ends at the missing premises. It does not recruit Schaffer's conclusion, and it does not claim to have blocked it.

What remains is a diagnostic use of his stereo, not a monistic conclusion. The stereo says: name the knobs. If the knobs are operation types, the three-workshop model sets them freely and then constrains a different variable, the history. If the knobs are event values inside one substitution, the model does not set them freely, and neither does a copying circuit. If the knobs are intrinsic natures of concrete parts, the model has not spoken. Priority monism is a further thesis about which of those knobs, if any, are the dependent fragments of a prior whole. A cyclic Boolean rule is not that thesis.

5 Implementation

Chalmers (2011) gives an account of implementation that already refuses to read a computation off a single actual history. The paper was written in 1993. He reports that its implementation section appeared in Chalmers (1994), and the text consulted here is the version he later published without substantive revision apart from one added note. The account is not a *Mind* article, and its claims about minds are not the target.

A physical system implements a finite-state automaton, on his definition, when formal states are mapped to physical state-types so that every formal transition $(S, I) \rightarrow (S', O')$ is matched by a reliable causal transition: if the system is in the physical state and receives the physical input that map to S and I, this reliably causes the state and output that map to S' and O'. The conditional is required for every transition in the specification, not only for the transitions that occur on one run. Combinatorial-state automata strengthen the partition. An internal state is a vector of components, naturally corresponding to distinct physical regions, and the causal transitions must mirror the formal transitions of the vector. Finite combinatorial automata are no more powerful, in the computability sense, than finite-state automata. Their implementation conditions are finer, because they demand organized causal interaction among parts. Chalmers's 2011 note, added to keep the view compatible with his views on consciousness, says that computational sufficiency for mentality is to be read as nomological rather than metaphysical. That note is recorded so that no metaphysical sufficiency claim is borrowed from him. Consciousness is not at issue here.

The three-workshop model does not refute this account. It fails to satisfy it until the missing physical clauses are filled in. Chalmers asks for reliable causation under the physical states that map to every formal transition. The product F is a set of formal functions. Rewriting a function

in a definition is not yet putting a physical component into a state. A whole-system serial simulation makes the gap vivid. A programmer can list the eight output triples, test $o = f(w(o))$, and print the unique solution. That procedure implements a search, if it implements anything. The search has a first step and an ordinary causal organization: read a line, test a predicate, write a bit. It does not follow that three physical regions, corresponding to the three slots, causally interact as the combinatorial automaton of the network would require. Ordered simulation and slot-level organization are different explanatory descriptions. Chalmers's own distinction between implementing a complex computation and implementing some simpler computation that merely shares an input-output table is the right distinction to use. Return to the same three workshops, still as a mathematical network. Suppose a programmer, who is not a fourth workshop in the network, lists the eight output triples and tests the fixed-point equation. The programmer's activity has parts: a list, a test, a printed triple. Those parts can implement a search in Chalmers's sense if the physical transitions of the programmer's machine reliably match the search. The printed triple may equal the unique history of some f. Equality of that triple is an input-output agreement between the search and the network. It is not a mapping under which each workshop is a physical region whose state reliably causes the next state of the vector (a, b, c) according to w. One can describe the programmer's machine without mentioning three slots. One can describe the three slots without mentioning the programmer. Clause (i) of the principle is the demand to say which of these partitions the implementation claim is about. A claim that slides from one to the other mid-sentence has not been assessed.

Sharing the printed record is not sharing the organization.

Nor does the model show that his counterfactual requirement was unnecessary. The requirement is what blocks the slide from one manifested history to implementation. The process-function quantifier is a formal cousin of that requirement: every substitution, not only one story, must have a matching history. The cousin lives in the space of functions. His requirement lives in the space of physical state-types and reliable causes. The typed substitution principle says that an implementation claim has to locate itself in one of these spaces explicitly.

The principle has three clauses.

(i) Partition. Say which components are in the claim, and which operation types on those components are being held fixed or varied.

(ii) Admission. Say which combinations of those substitutions are included. For the workshops, the included set is the full product F, or, in the reversible test, the full class of matched local unitaries.

(iii) Correspondence. Say which map from formal histories to physical responses is asserted to be reliable, and under which interventions.

Clause (i) stops a theorist from treating "the system" as an unstructured whole when the formalism is a vector of slots, and equally stops a theorist from treating formal slots as physical parts before a decomposition is given. Clause (ii) stops the equating of operation independence with event recombination. Clause (iii) is where Chalmers's reliable-cause condition, or any similarly serious counterfactual or mechanistic condition, still has to be earned. Satisfying (i) and (ii) inside a formal model does not satisfy (iii).

Score the two invented cases against the three clauses, so the principle is a test rather than a slogan.

For the serial circuit, clause (i) names two gates and the wires between them. The operation types are the allowed settings of each gate. Clause (ii) admits every pair of settings. It does not admit every assignment of bits to the wires. Clause (iii) is empty in the vignette, because no physical device has been described. A real copying circuit in a laboratory could fill (iii) by saying which voltages correspond to which bits and which transitions are reliable. Until that is said, the vignette is a constraint on equations. It is still enough to show that history dependence is ordinary. It is not enough to show that anything computes.

For the three workshops, clause (i) names three slots and the four bit-functions on each. Clause (ii) admits every element of F, and, in the reversible test, every matched tuple of local unitaries. It admits exactly one history per substitution, not eight. Clause (iii) is again empty. The canonical unitary extension fills a formal robustness condition on M. It does not name a physical region, a voltage, or a reliable cause. Scoring the workshops as an implementation because they pass (ii) is the permissive mistake. Scoring them as metaphysically entangled because they fail event recombination is the restrictive mistake. The principle's only work is to make those two mistakes visible as failures to keep the clauses apart.

The same scorecard explains why a change of intervention class is not a change of the metaphysical subject. Bit-functions, stochastic mixtures of bit-functions, local unitaries on matched ancillas, and general instruments are four classes. A valid probabilistic process need only give total probability one under products of local stochastic maps. That is a fourth quantifier, over mixtures, not a new theory of concrete objects. The biconditional of Section 3 is only the unitary class, for the canonical U. An assessor who cites the bit-function product as if it were the unitary test, or the unitary test as if it were a report of experimental control, has changed clause (ii) without saying so. The principle requires the class to be re-stated whenever it changes. That requirement is bookkeeping. It is the bookkeeping that keeps a formal result from being graded as a laboratory result or as a recombination result.

Chalmers notes that a system which implements a complex computation will typically implement many simpler ones as well, and that there need be no single canonical computation "the" system is performing. He treats that limited interest-relativity as harmless, because it

remains an objective question whether a given system implements a given computation. The scorecard agrees, and it uses the point. The programmer's machine may implement the search. The workshops, were they built, might implement the network. A box that merely prints $(0, 0, 0)$ implements neither until a mapping and a class of transitions are specified. Interest can choose which claim is being made. It cannot make a printout into a three-slot causal organization, and it cannot make a fixed-point constraint into a failure of free recombination. Objectivity of each claim, once the three clauses are filled in, is compatible with there being many claims.

The principle is not a sufficient analysis of computing. A claim can meet all three clauses and still be false, if the cited causal transitions do not in fact hold. A claim can concern a system that computes some other function, as Chalmers notes that real machines do many computations at once, without the question "does it implement this one?" becoming non-objective. The principle is a filter on assessment. It is compatible with his account. It adds an instruction for candidates whose natural description is a product of local operations constrained by a global history equation: do not grade that description as if the product were a recombination of concrete natures, and do not grade the history equation as if it were already the causal mirror.

Finite search does not collapse the filter. Given an explicit table and an explicit f, the eight candidates can be checked. The example introduces no new computable function and no speed-up. Searchability shows that history dependence is not uncomputability. It does not show that the searcher realizes the slot-level architecture. Lack of that architecture does not show that the table is uncomputable. Both mistakes ignore clause (i).

6 Objections

The first objection says that the distinction is only a distinction of variables, and that a distinction of variables is not a philosophical result. The logical point is elementary, and it should be granted. The philosophical point is the pair of invalid inferences the variables make available once "independence" is used both for what may be installed and for what may happen.

From operation independence one cannot infer event recombination. The premises $(\forall f \in F)\, f$ is admitted and $(\forall f)\,|H(w, f)| = 1$ do not entail $(\forall f)(\forall o \in O)\, o \in H(w, f)$. The serial circuit is a countermodel with no cyclic glamour: every gate setting is admitted, and not every wire assignment is a solution. From history dependence one cannot infer that the local operations fail to be independent, and one cannot infer that concrete objects fail Schaffer's modal freedom. The premise $|H(w, f)| = 1$ does not mention natures, locations, or existence. Both inferences are invalid before any theorem about monism is reached. If they were the only errors in the neighbourhood, the result would be a small one. They are the errors that appear as soon as "independence" is used once for the product F and again for the set O.

The second objection says that independent choice of operations already presupposes physically independent components, so the formal product is doing metaphysical work. A modelling domain is not a physical ability. The theorem, such as it is, says that the theory's domain is the full product of function sets. It does not say that three laboratories can be manipulated independently, that their parts are ontologically independent, or that an experimenter has a knob. A realization claim needs evidence for the physical ability. Operation independence in the model is clause (ii) without clause (iii). Treating it as ontological independence of the workshops adds the bridge premises listed in Section 4 and does not derive them.

The third objection says that Chalmers already handles the issue, so the principle is idle. His counterfactual requirement is conceded, and it is not re-discovered here. What the principle adds is a demand to declare the partition and the intervention class when the candidate is not an ordinary automaton specification but a product of local rules with a global fixed-point constraint. Whole-system search and local-slot organization can realize different computations while agreeing on a printed triple. His account explains why that difference matters. The principle tells the assessor not to lose it when the word "independence" is applied to the product of rules. Idle repetition would be the claim that reliable causation is a new idea. That claim is not made.

The fourth objection says that every finite table is searchable, so the example is an ordinary computation and the metaphysical fuss is unearned. Searchability is granted. Enumerate output triples, test the equation, return the unique solution. No complexity class, no hypercomputation, and no violation of an ordinary bound on mechanical procedure is asserted. The fuss, which is not metaphysical extravagance, is the identification of that search with the network. An ordered simulator computes a result. It need not instantiate the component-level architecture the network describes. The absence of that architecture does not make the result uncomputable. The objection is right that the example is computationally ordinary. It is wrong if it concludes that ordinary computability answers the implementation question.

The fifth objection says that acyclic cases have dependent histories too, so the workshops show nothing, or, in the other direction, that every constrained circuit is a failure of recombination and a step toward monism. The control case is the reply to both directions. Dependent wire values are the daily bread of circuit equations. They are not nonseparability and not priority monism. The workshops' distinctive formal feature is universal consistency without a fixed order of slots. The reversible test shows that this feature is not vetoed by the canonical unitary insertions. That blocks one overstrong quantum filter on the formal composition. It does not promote the circuit, cyclic or not, into a metaphysical unity.

The sixth objection says that freedom of intervention is illusory, because consistency selects the outcomes, so the parties were never free to choose. Operational availability and the global

solution count are separate quantifiers. Every $f \in F$ is available in the model. For each such f, exactly one o solves the equation. The selection of o is not a retraction of f. It is the history dependence already stated. Nothing in that pair of quantifiers is a theory of libertarian agency, a semantics for counterfactuals in terms of nearest worlds, or a report of experimental control. Formal quantification is not a hand on a dial. The temptation is to hear "every local function is allowed" as "each workshop can do otherwise," in a sense that already includes agency or experimental control, and then to hear "only one history occurs" as a retraction of that ability. The retraction would be real if the second quantifier ranged over F and struck members out. It ranges over O, inside a fixed f. The technician's formal menu remains the whole of F_k. What is fixed is which output triple solves the menu once the menu choices are made. A theory of ability, if one is wanted, has to say whether ability attaches to the menu or to the solution. The network does not say. Importing a libertarian reading, or a nearest-world semantics, would be a further theory of the words "can" and "would." Those words are not given that theory by the symbol $\forall f \in F$.

The seventh objection says that the cyclic dependence proves, or refutes, a grounding circle. Write the would-be inference so that the missing step is visible. Premise: the input at slot 1 is a function of the outputs at slots 2 and 3, and similarly around the table. Conclusion: the concrete workshop at slot 1 ontologically depends on the workshop at slot 2, which ontologically depends on the workshop at slot 1, so grounding is cyclic, or so priority monism is true, or so it is false. The premise is about a function on bit values. Schaffer's relation D is a relation of ontological dependence for nature and existence, stipulated to be irreflexive and well-founded in the version of the argument he proves. The functional graph of w is not that relation. He distinguishes that relation from causation. A fortiori it is not established by a Boolean formula. An application to grounding would need the bridge premises of Section 4, including well-foundedness, which a cycle of functional dependence may fail even when a priority order among concrete objects is perfectly well-founded, or the reverse. No such application is claimed. The objection is right to demand the distinction. It is wrong to treat the distinction as already a theorem of the network.

7 Conclusion

The three workshops admit every combination of local bit-functions and, for each combination, a single history. An ordinary serial circuit already shows that the second clause is not metaphysical news. The news, which is modest, is that these two clauses are compatible with each other and with the absence of a fixed order among the slots, and that a canonical reversible extension does not disturb the compatibility or create an implementation. Schaffer's modal freedom concerns ways concrete things can be. Chalmers's implementation condition concerns

reliable causal transitions under a stated mapping. Neither condition is settled by the product of function sets, and neither is refuted by the fixed-point constraint.

The numbers to remember are small. Sixty-four substitutions, eight candidate histories, and, if the standing example holds, sixty-four solutions rather than five hundred and twelve. The sixty-four are the freedom of operations. The shortfall from five hundred and twelve is the dependence of histories. Neither number is a census of concrete objects or of possible worlds. The clauses to remember are the three in the typed substitution principle: name the partition, name the admitted combinations, and name the physical correspondence. The first two can be filled in for a formalism. The third is what an implementation claim still owes. Until it is owed, there is nothing to refute in monism, grounding, or computationalism, and nothing to announce as a built machine. The serial circuit remains the control that keeps the shortfall from five hundred and twelve from looking like a discovery about the cosmos. Copying gates already produce that kind of shortfall. The workshops add a full product of operations without a fixed slot order, and a reversible test that does not remove them from the class of process functions. That is the whole result. A further claim about minds, substances, or built devices is a premise these equations do not contain, and this paper does not add it. The assessment is finished when the clauses are separated. Readers who want a further metaphysics must bring their own bridge.

The typed substitution principle is the moral worth keeping. Say what varies, say which combinations of those variations are admitted, and say which physical responses are supposed to track the formal histories. Until the third clause is filled in, the network is a constraint on a formalism. It is not a verdict about monism, grounding, computationalism, or the existence of a machine.

8 Assistance

Preparation of this draft, including the organization of the argument and the wording of the appendix, was assisted by Grok Bot, a Cursor cloud agent. The assistance included generative composition, not only copy editing. No version identifier for the tool is recorded here, and none is invented. The assistant is not an author. Whether this section is the placement a journal requires at initial submission, as against a cover letter or an acknowledgements section omitted from anonymous review, has not been settled. Author verification of the draft is pending.

A The finite clauses

Parties k have finite nonempty alphabets I_k and O_k. Substitutions are tuples of functions $f_k : I_k \to O_k$. A rule w is a process function when every substitution has exactly one fixed point of $o \mapsto (f_k(w_k(o)))_k$. For every rule w,

$$U|o\rangle|e\rangle = |w(o) + e\rangle|o\rangle$$

is a basis permutation. Given unitaries V_k from $I_k \otimes X_k$ to $O_k \otimes Y_k$ with $|I_k|\,|X_k| = |O_k|\,|Y_k|$, the closed loop has entries

$$M[(s, y), (e, x)] = \prod_k V_k[(s_k, y_k), (w_k(s) + e_k, x_k)].$$

Exclusivity, if w is a process function and $s \neq s'$, yields a party with unequal outputs and equal inputs, and that factor of $MM^\dagger$ vanishes. The converse gadget $V_k|i, x\rangle = |f_k(i) + x, i\rangle$ makes a column norm equal the number of fixed points. So M is unitary for every such family if and only if w is a process function. The biconditional is for this U and these tests. Proofs are in the unpublished note. They were not re-checked by a proof assistant for this draft.

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